SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

1. Smart Surveillance System Selection
A city surveillance system must operate under the following conditions:
- If latency > 100 ms → system fails
- If accuracy < 85% → unacceptable
- Hardware supports only moderate computation

Models available:
Viola-Jones: 78%, low latency
YOLO: 92%, moderate latency
Deep Learning: 95%, high latency

Compare the models and evaluate which satisfies all constraints. Justify your decision using logical reasoning based on given conditions.

ANSWER:

Requirements & Constraints: Latency ≤ 100 ms, Accuracy $\geq 85\%$, Hardware: Moderate computation.

Model Evaluation:

- *Viola-Jones:* Accuracy = 78% ($< 85\%$). Fails accuracy requirement.
- *Deep Learning:* Accuracy = 95%, but high latency (> 100 ms) and heavy compute load beyond moderate hardware. Fails latency and hardware constraints.
- *YOLO:* Accuracy = 92% ($\geq 85\%$), moderate latency (≤ 100 ms), supported by moderate hardware.

Decision: Select **YOLO**.

Justification: YOLO is the only model that simultaneously satisfies all three constraints: it meets the accuracy threshold ($92\% > 85\%$), operates within latency limits, and fits moderate hardware capabilities.

Model	Accuracy	Latency	Hardware Requirements	Constraint Verification	Decision Status
Viola-Jones	78%	Low ($\leq$ 100 ms)	Low compute	Accuracy $<$ 85%	**Rejected** (Violates accuracy floor)
YOLO	**92%**	**Moderate ($\leq$ 100 ms)**	**Moderate compute**	**All conditions satisfied**	**Selected**
Deep Learning	95%	High ($>$ 100 ms)	High compute	Latency $>$ 100 ms & compute overload	**Rejected** (Violates latency & hardware constraints)

3. Motion Analysis Decision Problem

A traffic system requires:

- Accurate motion direction detection
- Robustness in dense traffic

Results:

Optical Flow: accurate but noisy in dense scenes

Motion Estimation: stable but less detailed

Compare both approaches and evaluate which is suitable under high-density conditions. Justify using scenario-based logic.

ANSWER:

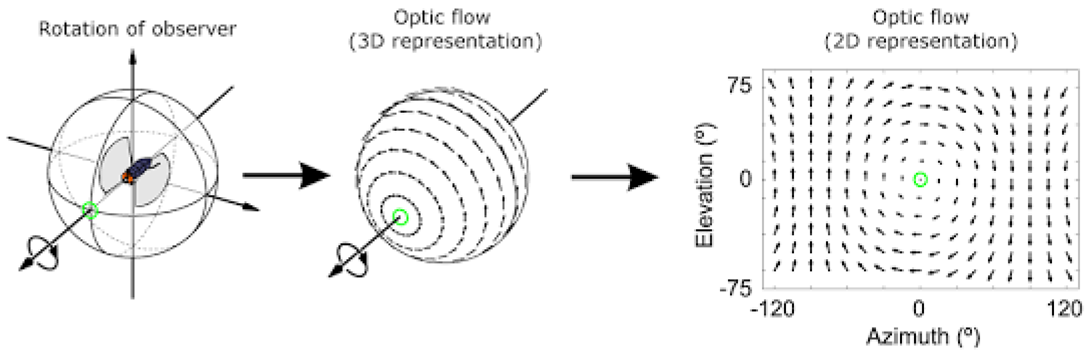
Requirements & Constraints: Accurate motion direction detection, robustness in high-density traffic.

Method Evaluation:

- *Optical Flow:* Highly accurate for fine motion vectors, but sensitive to occlusion, aperture problems, and noise in dense scenes.
- *Motion Estimation:* Provides region-based or block-based tracking. Less detailed, but significantly more stable against noise and pixel-level clutter.

Decision: Select **Motion Estimation** (or a hybrid approach using Motion Estimation for general tracking).

Justification: Under high-density conditions, pixel-level optical flow suffers from severe noise and trajectory overlap due to vehicle occlusions. Motion Estimation maintains structural stability and direction detection despite losing minor visual details.



4. Background Modeling Evaluation

A GMM-based system produces:

- Static scenes: 90% accuracy
- Dynamic scenes: 65% accuracy

Requirement:

System must maintain ≥80% accuracy across all scenarios

Compare performance and evaluate whether GMM alone meets requirements. Justify logically and suggest a decision.

ANSWER:

Requirements & Constraints: Maintain $\geq 80\%$ accuracy across all scenarios (static and dynamic).

Performance Evaluation:

- *Static Scenes:* 90% accuracy ($\geq 80\%$).
- *Dynamic Scenes:* 65% accuracy ($< 80\%$).
- *Overall:* GMM alone fails the requirement because it cannot handle dynamic background variations (e.g., swaying trees, water ripple, sudden illumination changes).

Decision: GMM alone does NOT meet requirements.

Suggested Enhancement: Implement Adaptive GMM with multi-scale texture features (e.g., LBP) or integrate a spatio-temporal filter (e.g., optical flow / deep background subtraction) to dynamically adjust learning rates for unstable background pixels.

9. Mobile Face Detection Trade-off

Constraints:

- Real-time operation required
- Accuracy ≥85%

Options:

Viola-Jones: 80%, fast

Deep Learning: 95%, slow

Compare and evaluate which approach should be chosen. Justify using constraint satisfaction logic.

ANSWER:

Requirements & Constraints: Real-time operation required, Accuracy $\geq 85\%$.

Option Evaluation:

- *Viola-Jones:* Fast (real-time), but Accuracy = 80% ($< 85\%$). Fails accuracy constraint.
- *Deep Learning:* Accuracy = 95% ($\geq 85\%$), but slow (violates real-time requirement on standard mobile hardware). Fails latency constraint.

Decision: Neither off-the-shelf option meets all constraints simultaneously.

Justification: Viola-Jones violates accuracy thresholds; standard Deep Learning violates real-time execution bounds.

Recommended Engineering Solution: Deploy a **Lightweight/Quantized Deep Learning Model** (e.g., MobileNet-SSD, BlazeFace, or ONNX-quantized model) that yields $\geq 85\%$ accuracy while achieving real-time speeds on mobile NPUs/GPUs.

14. Motion Tracking in Sports Analytics

Requirements:

- Accurate motion tracking
- Real-time performance

Options:
 Optical Flow: accurate, slow
 Motion Estimation: fast, less accurate

Compare and evaluate the suitable approach. Justify using constraints.

ANSWER:

Requirements & Constraints: Accurate motion tracking, real-time performance**.**

Option Evaluation:

- *Optical Flow:* High accuracy, but computationally heavy and slow (violates real-time constraint).

- *Motion Estimation:* Fast (meets real-time requirement), but lower accuracy.

Decision: Select Motion Estimation (or optimized sparse Optical Flow).

Justification: Real-time response is a hard constraint for live sports analytics. Motion Estimation satisfies processing time limits while maintaining acceptable directional tracking. If high accuracy is mandatory, using sparse Optical Flow (e.g., Lucas-Kanade on key features) bridges the gap between speed and accuracy.

22. Edge Device Vision System

Constraints:

- Low power
- Real-time performance

Options:
 Deep Learning: accurate, heavy
 Lightweight Model: moderate accuracy, efficient

Compare and evaluate. Justify logically

ANSWER:.

Requirements & Constraints: Low power consumption, real-time processing, deployment on edge hardware.

Option Evaluation:

- *Deep Learning (Standard):* High accuracy, but heavy compute requirements (> 10 W), high thermal output, and non-real-time execution on low-power hardware.
- *Lightweight Model:* Moderate accuracy, high computational efficiency, low power usage, real-time response.

Decision: Select Lightweight Model.

Justification: Edge deployments are bound strictly by thermal and power budgets. A lightweight model fits within thermal/power limits and delivers real-time inferences, making it the viable deployment choice.

24. Retail Analytics System

Requirements:

- Customer tracking
- Moderate accuracy
- Real-time processing

Options:
Method A: accurate, slow
Method B: moderate accuracy, fast

Compare and evaluate. Justify your decision.

ANSWER:

Requirements & Constraints: Customer tracking, moderate accuracy, real-time processing.

Option Evaluation:

- *Method A:* High accuracy, slow processing speed. Fails real-time requirement.
- *Method B:* Moderate accuracy, fast processing speed. Meets both accuracy and speed bounds.

Decision: Select Method B.

Justification: Retail analytics (e.g., heatmapping, foot-traffic counts) explicitly accepts moderate accuracy. Method B meets the exact baseline requirement while preserving real-time operational flow.

25. Object Detection in Foggy Conditions

Requirements:

- Robust under poor visibility
- Accuracy ≥ 85%

Options:
Model A: 80%
Model B: 87%
Model C: 90% but slow

Compare and evaluate. Justify your decision.

ANSWER:

Requirements & Constraints: Robust under poor visibility, Accuracy $\geq 85\%$.

Option Evaluation:

- *Model A:* Accuracy = 80% ($< 85\%$). Fails baseline accuracy constraint.
- *Model C:* Accuracy = 90%, but slow. Impractically slow for real-world visibility applications.
- *Model B:* Accuracy = 87% ($\geq 85\%$), robust performance.

Decision: Select **Model B**.

Justification: Model B is the optimal trade-off; it surpasses the 85% accuracy floor (87%) without incurring the high latency penalty of Model C.

Code of Conduct:

I ______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation